

# Design of Operant Chamber for Chronic Brain Injury Research

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## INTRODUCTION

- To support discoveries of how traumatic brain injuries influence social behavior, the OSU Chronic Brain Injury Behavioral Lab required a cost-effective way to administer testing.
- Currently on the market, there are no apparatuses available that support the specific need of a 3-chamber design with dividing doors and a food pellet dispenser.

## OUR ROLE

- As Multidisciplinary Capstone students, our role was to design and build an environment to enable the lab to test social behavior in rats through motivating factors.
- By creating a custom-designed chamber, we were able to directly meet the needs of the lab and rats at a far-reduced cost compared to commercially available units.
- Our chamber enables controlled, repeatable experiments that will contribute to neuroscience research.

## DESIGN DECISIONS

- The design requirements and user needs were identified through market analysis of existing products, as well as discussions with the client.
- Stainless steel was selected as the primary material due to its durability, longevity, cost-effectiveness, and availability. It provides a flat, cleanable, and antimicrobial surface for contact with the rats, while also resisting damage from chewing and scratching behaviors.
- Software was developed using an Arduino to control door actuation and sensor input, enabling automated and repeatable experiments.
- The fabrication plan was developed with the goal of the most efficient sequence of construction. The process began with building the outer frame to gain familiarity with materials, followed by the assembly of more complex components such as the door mechanism, dividing walls, and front plexiglass panels.

## DESIGN PROCESS & FABRICATION

### SolidWorks CAD Design

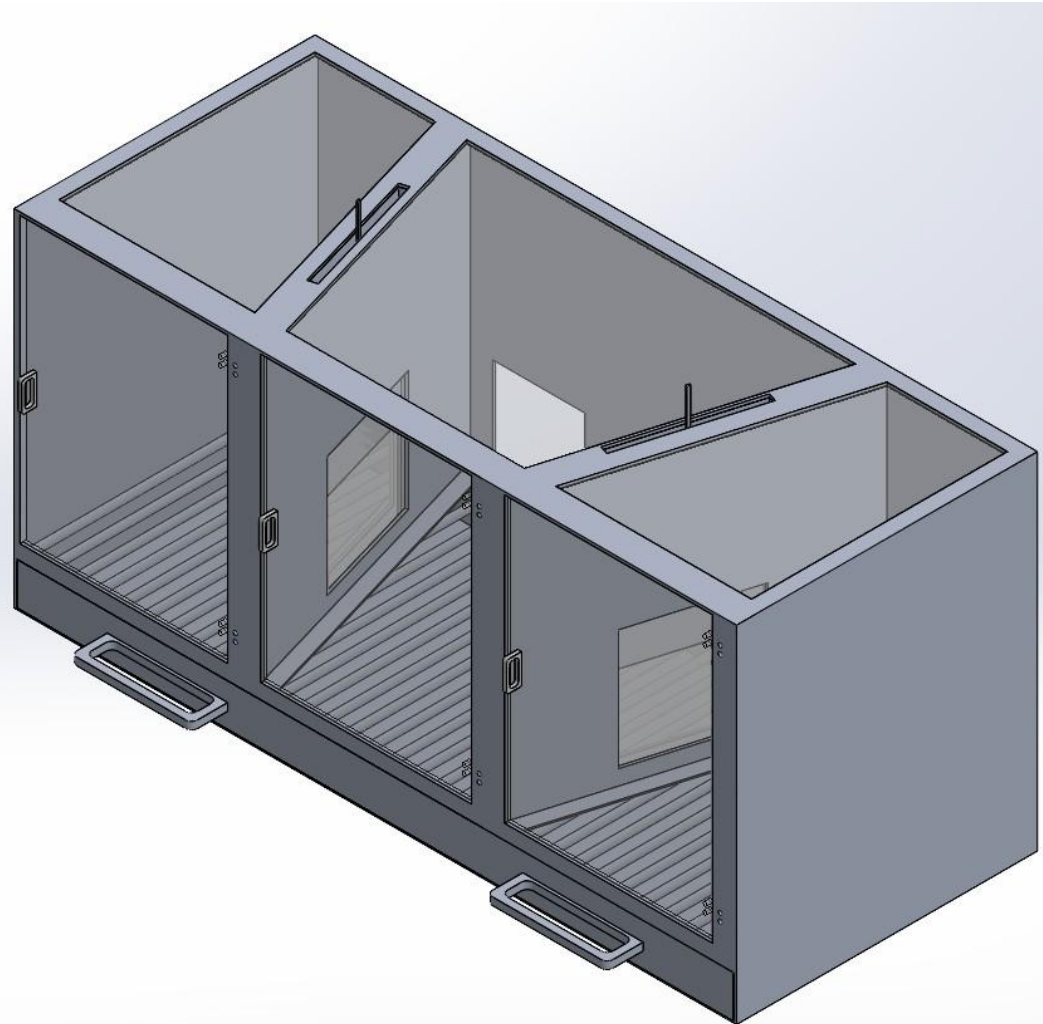


Figure #1 – Chamber Front

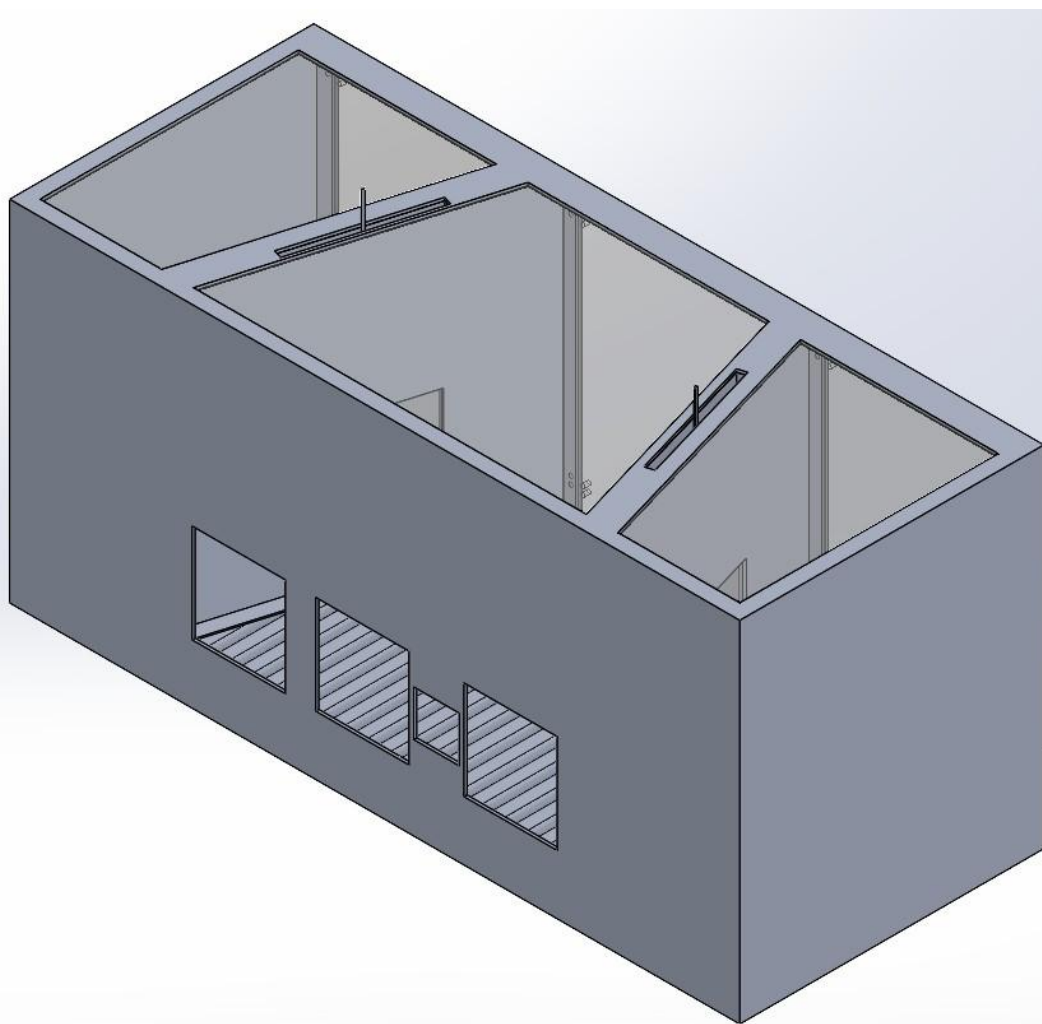


Figure #2 – Chamber Rear

- The chamber has two social rooms and one decision-making room, each containing one rat
- The center room features cutouts in the rear for three levers and one pellet tray.
- Each room is separated by a sliding door mechanism, which allows for sight, smell, and touch interaction between rats.

### Door Mechanism

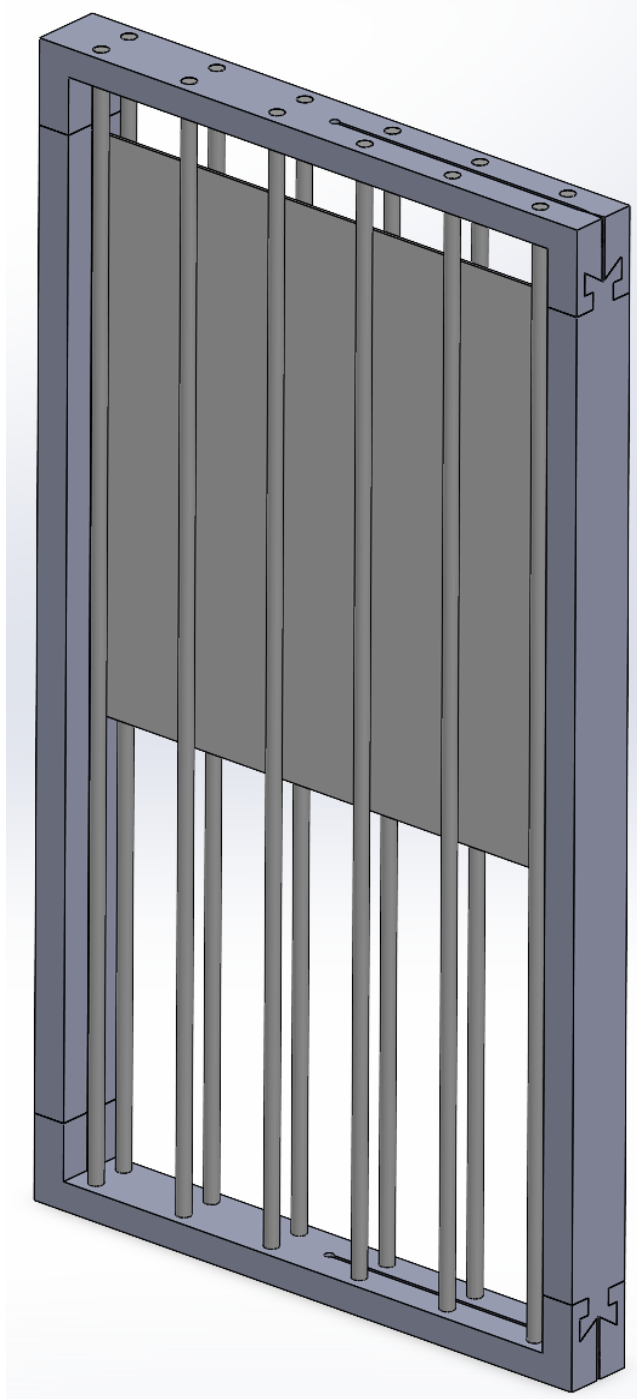


Figure #3 – Divider Wall

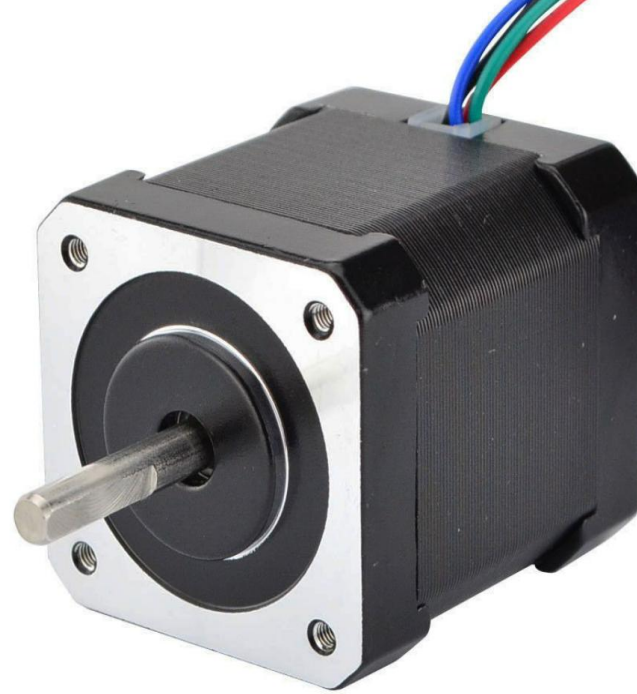


Figure #4 – Motor

- The doorway cartridge has separating bars to prevent cross chamber traversal.
- A steel door slides up and down within bars.
- The door is actuated by a rope on a motorized spool on the roof above the door slot.

### Final Chamber

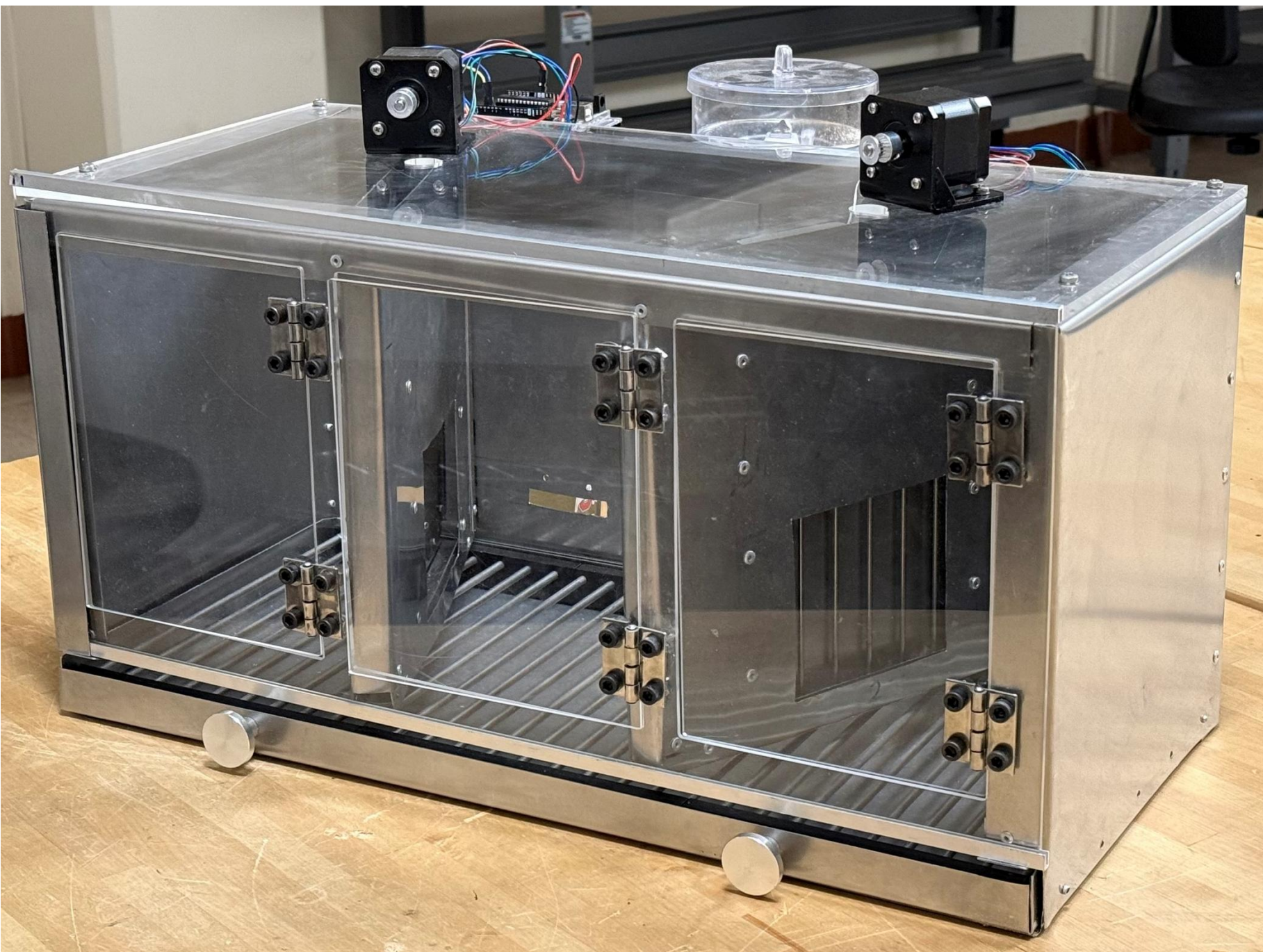


Figure #5 – Final Assembled Chamber

### Testing & Validation

| Category                    | Test                                                         | Purpose/Outcome                                           |
|-----------------------------|--------------------------------------------------------------|-----------------------------------------------------------|
| Motor Performance           | Run 100 trials to lift and lower the door.                   | Ensure motor can provide enough force at reasonable temp. |
| Ability to Clean Waste Tray | Slide tray out with <5lb force, remove grate, reinstall      | Confirm smooth tray motion and ease of dumping waste.     |
| Rat Safety                  | Apply pressure to all internal metal edges & fasteners       | Check for sharp edges and corners, reduce chew points.    |
| Maintenance & Modularity    | Remove roof and swap door cartridges with only a screwdriver | Support maintenance by reducing need for special tools    |

Figure #6 – Test Plan

### Fabrication Methods

- Sheet Metal:** Industrial shear to cut to size. Bent in a sheet metal brake. Riveted together using 1/8" rivets. Jigsaw used for door and lever cutouts.
- Plexiglass:** Cut door frames, holes, etc. using CNC Mill. Secured with bolts for lower compression.
- Rods:** Metal band saw used to cut to size, belt sander used to taper ends and debur.
- Rod Mounts for Door & Waste Tray:** 3D Printer used to achieve exact sizing and placement of rods.

## CONCLUSION

- The team designed and manufactured a three-room operant chamber to quantitatively assess social reward compared to sugar reward.
- Two intermediate doors controlled by a lever press were designed and installed to meet testing requirements. Software was developed to integrate sensors and motor control, allowing for automated and repeatable experiments.
- Budget: Our total cost (\$1,061.38) was significantly cheaper than most market options (\$3000+). This reflects our user needs of affordability and efficiency.
- Reproducibility: The CBI lab can use the current custom chamber as a reproducible artifact. It can either be duplicated for extension of TBI behavior testing on rats or can be scaled down for use with mice.

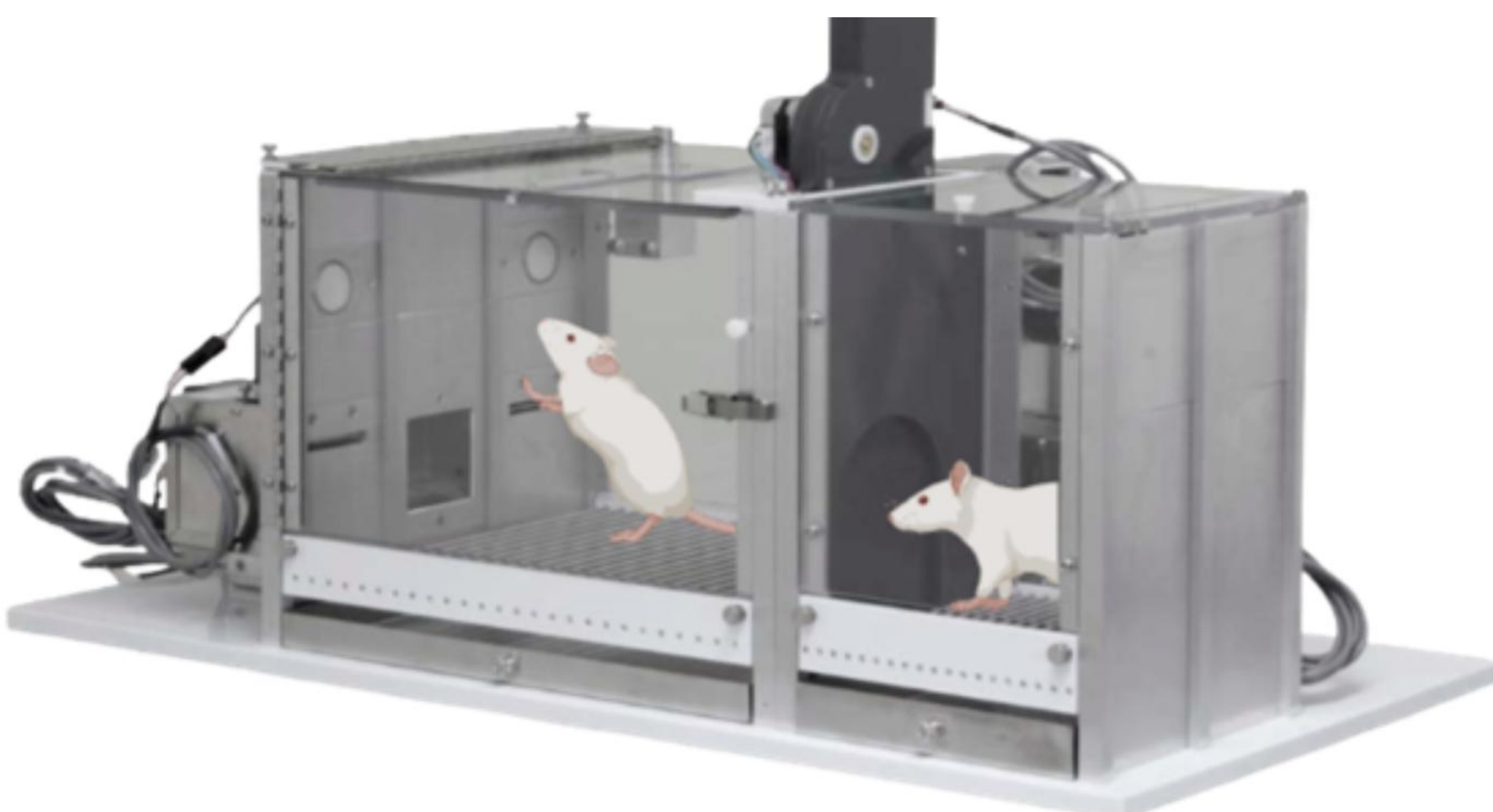


Figure #7 – Existing Market Option

## ACKNOWLEDGEMENTS

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